Murilo Pinto Alves Apparecido

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Education

University of Michigan Ann Arbor, MI

Bachelors in Computer Science, Minor in Mathematics *Expected May 2028*

Organizations: National Society of Black Engineers, ColorStack, Google Developers Student Club, Phi Beta Sigma, Phi Gamma Nu, Codepath

Technical Skills

Languages: Python, C/C++, JavaScript, TypeScript, Swift, Dart, Java, HTML/CSS
Frameworks & Tools: Jira, React Native, Flutter, Expo, Node.js, Firebase, Supabase, PostgreSQL, Git, Figma
Libraries & Data: Gurobi, NumPy, SciPy, GeoPandas, scikit-learn, Rasterio, Matplotlib, Google Earth Engine

Experience

Full-Stack Engineering Intern May 2026 – Present

Center for Healthcare Engineering & Patient Safety (CHEPS), University of Michigan *Ann Arbor, MI*

- Built a full-stack spreadsheet editor (React, TypeScript, Spring Boot, PostgreSQL) to manage input files for a Python/Gurobi MIP solver that generates optimal provider schedules for Michigan Medicine departments, replacing a manual Excel-based workflow in active clinical use since 2015.
- Designed a cross-file column linking system enabling shared data references across spreadsheets, reducing redundant input files which minimized duplicate constraint data passed to the Gurobi MIP model.
- Architected an optimistic update system with TanStack Query and a multi-state change tracker (create, update, delete) enabling batch saves across rows, columns, and cells with zero page reloads.
- Resolved stale closure bugs in memoized React column definitions, engineered a negative-ID temporary identifier system for tracking unsaved rows and columns client-side, and designed a batch create endpoint that reconciles temporary IDs with real database IDs server-side.

Undergraduate Research Assistant August 2025 – April 2026

University of Michigan *Ann Arbor, MI*

- Engineered a 60,701-cell geospatial risk surface over Chicago (100m resolution) integrating 2021 population density, building coverage ratios, and a ballistic debris model to quantify expected ground casualties per crash event.
- Implemented Theta* any-angle path planning with a blended risk-distance cost function, computing eVTOL corridors that achieve 60–80% casualty risk reduction at only 10–20% added flight distance.
- Developed a CA-ANN model forecasting population density and building coverage for 2030/2035, generating forward-looking risk surfaces and running Theta* on all three snapshots to compare route efficiency over time.

Projects

Codebase Q&A Agent | *Python, OpenAI API, ChromaDB, RAG*

- Built a retrieval-augmented generation system enabling natural language Q&A over a production codebase, ingesting and chunking 100+ files across TypeScript and Java.
- Diagnosed a semantic search blind spot where code identifiers (e.g. function names) failed to rank in top-20 retrieval results, and designed a hybrid search using vector similarity with keyword matching and score-based merging.
- Implemented an LLM-based query rewriting layer to bridge the semantic gap between natural language questions and code-specific terminology, improving retrieval relevance for test queries.

ViaShield Safety App | *React Native, Expo, Mapbox, Valhalla*

- Built the consumer-facing mobile app for a safety intelligence platform scoring geographic coordinates 1–10 for danger using crime data, NASA Black Marble radiance, and OSM environmental features across Bogotá, Medellín, São Paulo, and San Francisco.
- Implemented walking navigation by wiring Valhalla's pedestrian routing profile with pedestrian-specific safety scoring weights, expanding the platform beyond its existing driving-only routing.
- Architected the incident reporting flywheel, user-submitted crime reports requiring 2+ independent device confirmations, to grow the crowdsourced dataset feeding scoring for all B2B enterprise clients.